# Racial Differences in Adolescent Cyberbullying Victimization: The Moderating Role of Perceived School Safety

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## Introduction

- Pew Research Center (2022) reports that **Black and Hispanic teens are significantly more likely than their White peers** to say they think their race makes them a target of online abuse, including **cyberbullying**.
- Cyberbullying is defined as "an aggressive, intentional act or behavior through digital mediums, repeatedly and over time"
  - associated with depression, low self-esteem, behavioral problems, and substance abuse (Hamm et al, 2015).
- These racial differences may stem from **in-person discrimination** that becomes amplified in an **online environment** and requires further investigation to better understand the underlying mechanisms linking these two forms of victimization.
- While prior research has demonstrated that adolescents' **cyberbullying experiences are unevenly distributed across racial groups** (Alhajji et al., 2019), less is known about the factors that contribute to these disparities.
  - One possible explanation may be **safety concerns at school**, a factor previously associated with traditional bullying (Steiner & Rasberry, 2015), yet few studies have examined how such concerns may be associated with online victimization.





- The present study explores whether **safety concerns at school help account for racial differences in cyberbullying victimization**.
- Feelings of unsafety and school avoidance may reflect broader school climate factors and could be associated with differences in youths' online exposure and interactions.
- Results from this study offer parents and educators insights for designing targeted prevention strategies that address the needs of at-risk students and promote safer school climates across in-person and virtual settings.

## Research Question

(1) Are there differences in the likelihood of experiencing cyberbullying between White youth and youth from racial minority groups?
(2) Does the association between race and cyberbullying vary depending on perceived school safety concerns?

## Methods

**School Safety (Moderator):**
**In the past 30 days, how many days did you not go to school because you felt you would be unsafe or on your way to or from school?:**
1. 0 days; 2. 1 day; 3. 2 or 3 days; 4. 4 or 5 days; 5. 6 or more day

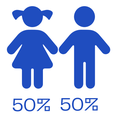
**Race/Ethnicity (IV):´**
**What is your race?:**
1. American Indian/ Alaska Native; 2. Asian; 3. Black or African American; 4. Native Hawaiian/ Pacific Islander; 5. White; 6. Multiracial; 7. Hispanic

**Cyberbullying (DV):**
**During the past 12 months, have you ever been electronically bullied?**
(Count being bullied through texting, Instagram, Facebook, or other Social Media.): 1. Yes; 2. No

### Statistical Analysis

- Logistic regression tested racial differences in cyberbullying victimization. (White = reference group).
- Moderation analysis tested whether school safety concerns (measured as days of school avoidance) influence this relationship.

### Participants




- 19,553 students from 311 schools
- 9th - 12th grade students in the USA (≈ 16 years old on average)

### Data Source

Data is from the public Youth Risk Behavior Surveillance System dataset (2023), a national survey conducted by the CDC (Centre for Disease Control and Protection).

## Results

**Table 1**
*Descriptive Statistics*

| Variable | *n* | *Frequency* | *Percentage* |
|---|---|---|---|
| **1. Race** | 19,740 | | |
| American Indian/ Alaska Native | | 1,334 | 6.6 |
| Asian | | 995 | 4.9 |
| Black/ African American | | 1,791 | 8.9 |
| Native Hawaiian/ Pacific Islander | | 105 | .5 |
| White | | 9,700 | 48.3 |
| Hispanic | | 3,997 | 19.9 |
| Multiracial | | 1,818 | 9.0 |
| **2. Cyberbullying** | 19,898 | | |
| Yes | | 3,423 | 17.0 |
| No | | 16,475 | 82.0 |
| **3. School safety concerns** | 18,982 | | |
| 0 days | | 16,681 | 83.0 |
| 1 day | | 1,094 | 5.4 |
| 2 or 3 days | | 692 | 3.4 |
| 4 or 5 days | | 183 | .9 |
| 6 or more days | | 332 | 1.7 |

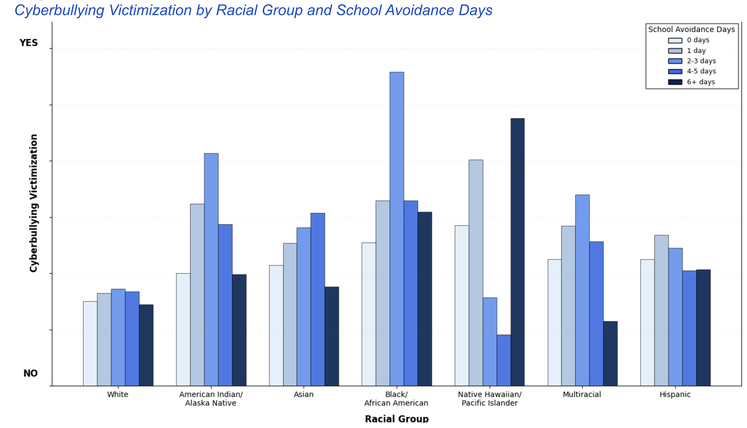
A binary logistic regression was conducted to examine the predictors of electronic bullying victimization. The final model was statistically significant, $\chi^2(34) = 951.86$, $p < .001$, explaining 1.1% of the variance (Nagelkerke $R^2 = .011$). Compared to White students, several racial minority groups had significantly higher odds of experiencing electronic bullying.

**Table 2**
*Odds Ratio of Cyberbullying Prevalence by Racial Group*

| Racial Group | β | OR | 95% CI | *p* |
|---|---|---|---|---|
| Black | **0.64** | **1.90** | **[1.58 – 2.28]** | **<.001** |
| Multiracial | -0.10 | 0.90 | [0.78 – 1.05] | .170 |
| Hispanic | **0.47** | **1.60** | **[1.41 – 1.81]** | **<.001** |
| Asian | **0.42** | **1.52** | **[1.22 – 1.89]** | **<.001** |
| American Indian/Alaska Native | **0.34** | **1.41** | **[1.14 – 1.74]** | **.001** |
| Native Hawaiian/Pacific Islander | 0.80 | 2.23 | [0.97 – 5.16] | .060 |

*Note. Reference group = White students. OR = odds ratio. CI = confidence interval. Bolded values are statistically significant at p < .05. All p values are two-tailed.*

**Graph 1**
*Cyberbullying Victimization by Racial Group and School Avoidance Days*




*Note. Bars represent the proportion of students reporting cyberbullying victimization (0 = No, 1 = Yes) by days of school avoidance due to safety concerns. Darker blue shades indicate more days of school avoidance. Data from Youth Risk Behavior Surveillance System (YRBS).*

The addition of School Safety Concerns significantly improved the model's predictive power $\chi^2(24) = 40.46$, $p = .019$.

- American Indian/Alaska Native (*OR* = 2.76, *p* = .002), Black (*OR* = 2.19, p = .003), and Multiracial students (OR = 1.51, *p* = .015) who missed 2–3 days of school due to feeling unsafe had significantly higher odds of experiencing electronic bullying
- Multiracial students (*OR* = 0.51, *p* = .042) who missed 6 or more days showed significantly lower odds, indicating a complex pattern of moderation.

## Discussion

### Key Findings

- Consistent with previous research, cyberbullying victimization is not evenly distributed across racial groups among U.S. adolescents.
  - African American youth were significantly more likely, followed by Hispanic, Asian, and American Indian/Alaska native youth, to report experiencing cyberbullying compared to White youth.
- School safety concerns partially moderated racial differences in cyberbullying, with the association between school avoidance and electronic bullying varying across groups.
  - American Indian/Alaska Native, Black, and Multiracial students who reported missing 2–3 days of school due to feeling unsafe showed significantly higher odds of experiencing electronic bullying
- Overall, the findings emphasize the importance of considering race when examining cyberbullying and suggest prevention efforts should address broader issues of discrimination and school climate that may shape youths' online interactions.
  - However, the variance suggests cyberbullying is a highly complex phenomenon influenced by a broad range of factors beyond school safety and race.

### Future Directions

- Future research can employ longitudinal designs to better understand the relationship between race, school safety concerns, and cyberbullying victimization
- Incorporate additional measures of school climate and safety (e.g. teacher support, peer norms, campus security) to provide a more comprehensive understanding of the influence of school environments
- Investigate whether similar racial disparities in cyberbullying are observed in other cultural or national contexts



## References

## Contact Us

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